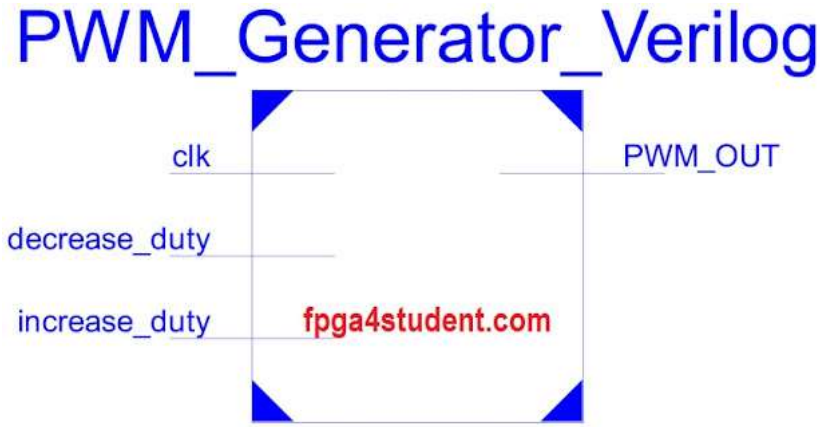
Verilog code for PWM generator

This Verilog project presents a Verilog code for PWM generator with Variable Duty Cycle. Last time, I presented a VHDL code for a PWM generator.
The Verilog PWM (Pulse Width Modulation) generator creates a 10MHz PWM signal with variable duty cycle. Two buttons which are debounced are used to control the duty cycle of the PWM signal. The first push button is to increase the duty cycle by 10%, and the other button is to decrease the duty cycle by 10%.



Verilog code for PWM generator with variable duty cycle:

```
// fpga4student.com: FPGA Projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for PWM Generator with variable Duty Cycle
// Two debounced buttons are used to control the duty cycle (step size: 10%)
module PWM_Generator_Verilog
 (
 clk, // 100MHz clock input
 increase_duty, // input to increase 10% duty cycle
 decrease_duty, // input to decrease 10% duty cycle
 PWM_OUT // 10MHz PWM output signal
    );
 input clk;
 input increase_duty;
 input decrease_duty;
 output PWM_OUT;
 wire slow_clk_enable; // slow clock enable signal for debouncing FFs
 reg[27:0] counter_debounce=0;// counter for creating slow clock enable signals
 wire tmp1,tmp2,duty_inc;// temporary flip-flop signals for debouncing the increas
 wire tmp3,tmp4,duty_dec;// temporary flip-flop signals for debouncing the decreas
 reg[3:0] counter_PWM=0;// counter for creating 10Mhz PWM signal
 reg[3:0] DUTY_CYCLE=5; // initial duty cycle is 50%
  // Debouncing 2 buttons for inc/dec duty cycle
  // Firstly generate slow clock enable for debouncing flip-flop (4Hz)
 always @(posedge clk)
 begin
   counter_debounce <= counter_debounce + 1;
   //if(counter_debounce>=25000000) then
   // for running on FPGA -- comment when running simulation
   if(counter_debounce>=1)
   // for running simulation -- comment when running on FPGA
    counter_debounce <= 0;
 end
 // assign slow_clk_enable = counter_debounce == 25000000 ?1:0;
 // for running on FPGA -- comment when running simulation
 assign slow_clk_enable = counter_debounce == 1 ?1:0;
 // for running simulation -- comment when running on FPGA
 // debouncing FFs for increasing button
```

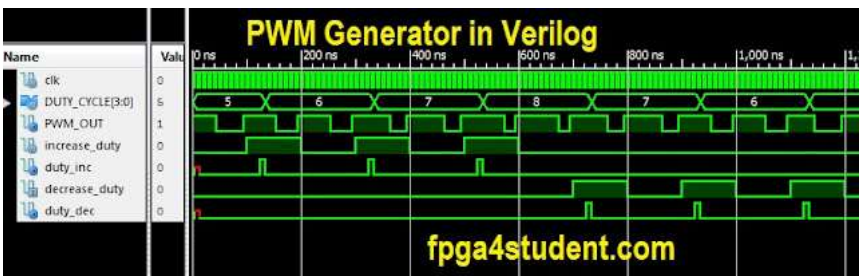
```
 DFF_PWM PWM_DFF1(clk,slow_clk_enable,increase_duty,tmp1);
 DFF_PWM PWM_DFF2(clk,slow_clk_enable,tmp1, tmp2);
 assign duty_inc =  tmp1 & (~ tmp2) & slow_clk_enable;
 // debouncing FFs for decreasing button
 DFF_PWM PWM_DFF3(clk,slow_clk_enable,decrease_duty, tmp3);
 DFF_PWM PWM_DFF4(clk,slow_clk_enable,tmp3, tmp4);
 assign duty_dec =  tmp3 & (~ tmp4) & slow_clk_enable;
 // vary the duty cycle using the debounced buttons above
 always @(posedge clk)
 begin
   if(duty_inc==1 && DUTY_CYCLE <= 9)
    DUTY_CYCLE <= DUTY_CYCLE + 1;// increase duty cycle by 10%
   else if(duty_dec==1 && DUTY_CYCLE>=1)
    DUTY_CYCLE <= DUTY_CYCLE - 1;//decrease duty cycle by 10%
 end
// Create 10MHz PWM signal with variable duty cycle controlled by 2 buttons
 always @(posedge clk)
 begin
   counter_PWM <= counter_PWM + 1;
   if(counter_PWM>=9)
    counter_PWM <= 0;
 end
 assign PWM_OUT = counter_PWM < DUTY_CYCLE ? 1:0;
endmodule
// Debouncing DFFs for push buttons on FPGA
module DFF_PWM(clk,en,D,Q);
input clk,en,D;
output reg Q;
always @(posedge clk)
begin
 if(en==1) // slow clock enable signal
  Q <= D;
end
endmodule
```

Verilog Testbench code for PWM generator:

```
`timescale 1ns / 1ps
// fpga4student.com: FPGA Projects, Verilog projects, VHDL projects
// Verilog project: Verilog testbench code for PWM Generator with variable duty cy
module tb_PWM_Generator_Verilog;
 // Inputs
 reg clk;
 reg increase_duty;
 reg decrease_duty;
 // Outputs
 wire PWM_OUT;
 // Instantiate the PWM Generator with variable duty cycle in Verilog
 PWM_Generator_Verilog PWM_Generator_Unit(
  .clk(clk),
  .increase_duty(increase_duty),
  .decrease_duty(decrease_duty),
  .PWM_OUT(PWM_OUT)
 );
 // Create 100Mhz clock
 initial begin
 clk = 0;
 forever #5 clk = ~clk;
 end
 initial begin
  increase_duty = 0;
  decrease_duty = 0;
  #100;
    increase_duty = 1;
```

```
  #100;// increase duty cycle by 10%
    increase_duty = 0;
  #100;
    increase_duty = 1;
  #100;// increase duty cycle by 10%
    increase_duty = 0;
  #100;
    increase_duty = 1;
  #100;// increase duty cycle by 10%
    increase_duty = 0;
  #100;
    decrease_duty = 1;
  #100;//decrease duty cycle by 10%
    decrease_duty = 0;
  #100;
    decrease_duty = 1;
  #100;//decrease duty cycle by 10%
    decrease_duty = 0;
  #100;
    decrease_duty = 1;
  #100;//decrease duty cycle by 10%
    decrease_duty = 0;
 end
endmodule
```

Simulation Waveform for PWM generator in VHDL:



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